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QuaRK: A Quantum Reservoir Kernel for Time Series Learning

 (/pdf?id=KxcW2BjzaH)

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 Research Track Cycle 2, Senior Area Chairs, Area Chairs, Reviewers, Authors

 Revisions (/revisions?id=KxcW2BjzaH)  BibTeX

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Keywords: Quantum Reservoir Computing, Quantum Kernels, Time Series

TL;DR: We propose QuaRK, a scalable quantum-classical method that turns windowed time series into quantum-reservoir features and learns a kernel readout with generalization guarantees on dependent data.

Abstract:

Quantum reservoir computing offers a promising route for time series learning by modelling sequential data via rich quantum dynamics while the only training required happens at the level of a lightweight classical readout. However, studies featuring efficient and implementable quantum reservoir architectures along with model learning guarantees remain scarce in the literature. To close this gap, we introduce \textsc{QuaRK}, an end-to-end framework that couples a hardware-realistic quantum reservoir featurizer with a kernel-based readout scheme. Given a sequence of sample points, the reservoir injects the points one after the other to yield a compact feature vector from efficiently measured k -local observables using classical shadow tomography, after which a classical kernel-based readout learns the target mapping with explicit regularization and fast optimization. The resulting pipeline exposes clear computational knobs—circuit width and depth as well as the measurement budget—while preserving the flexibility of kernel methods to model nonlinear temporal functionals and being scalable to high-dimensional data. We further provide learning-theoretic generalization guarantees for dependent temporal data, linking design and resource choices to finite-sample performance, thereby offering principled guidance for building reliable temporal learners. Empirical experiments validate \textsc{QuaRK} and illustrate the predicted interpolation and generalization behaviours on synthetic β -mixing time series tasks.

Corresponding Author:  Abdallah Aaraba (/profile?id=~Abdallah_Aaraba1)

Resubmission Flag: No

Artifact Pledge: Yes

Submission Guidelines Blind: Yes

Submission Guidelines Format: Yes

Submission Guidelines Limit: Yes

Submission Guidelines Authorship: Yes

Student Author:  Yes

Serve As Reviewer:  Shengrui Wang (/profile?id=~Shengrui_Wang2)

Reviewer Qualification:  N.A. - A qualified author is provided in the previous question

Information Sharing Consent:  Yes

Submission Number: 3103

Discussion (?id=KxcW2BjzaH#discussion)

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Paper Decision

Decision by Program Chairs 📅 15 May 2026, 22:15 (modified: 15 May 2026, 22:30) 👁 Program Chairs, Authors

📄 Revisions (/revisions?id=XmZ6lnS0Yo)

Decision: Reject

Comment:

This paper proposes QuaRK, which combines quantum reservoir computing with kernel ridge regression for time-series learning, and provides a theoretical analysis of generalization based on the Johnson–Lindenstrauss (JL) projection and β -mixing dependent data. The work demonstrates relatively strong performance in theoretical modeling, but shows significant weaknesses in empirical validation and practical relevance.

Strengths:

1. Proposes a unified quantum reservoir + kernel learning framework.
2. Provides relatively complete theoretical analysis, including:
 - Interpolation guarantees under JL projection.
 - Generalization bounds for β -mixing dependent data.
3. The overall architecture is modular and easy to understand.
4. Qualitatively validates theoretical trends on synthetic data.

Weaknesses:

1. Severely limited empirical evaluation:
 - Only low-dimensional synthetic VARMA datasets are used, with no real-world time-series datasets, making practical relevance unclear.
 - Lacks comparison against strong baselines.
2. Lack of ablation and statistical analysis:
 - The contribution of the JL projection is unclear.
 - No error bars or confidence intervals are reported.
 - No systematic hyperparameter sensitivity analysis is provided.
2. The scalability claims based on JL projection are not empirically supported.
3. The independent contribution of the quantum reservoir is not clearly demonstrated.
4. Real quantum noise and finite-shot effects are not considered.

The three reviewers present highly consistent opinions, all pointing out issues such as insufficient experiments, missing baselines, and unclear contributions of the quantum module. In summary, this is not a complete and well-substantiated paper in its current form, and it does not meet the standard required for acceptance at KDD.

Official Review of Submission3103 by Reviewer kc2J



Paper Summary:

This paper addresses the quantum reservoir problem. To overcome the limitations of quantum reservoir architectures, it proposes QuaRK, an end-to-end framework that couples a hardware-realistic quantum reservoir featurizer with a kernel-based readout scheme. The experimental results also validate the theoretical findings.

Paper Strengths:

1. This paper primarily proposes a framework called QuaRK and implements its extension.
2. It introduces the Johnson–Lindenstrauss (JL) projection method to form a new representation that matches the number of qubits in the selected quantum system.
3. Based on relevant time-series data, it provides learning-theoretic guarantees, linking design/resource selection to finite-sample performance, thereby combining theory with practice.

Paper Weaknesses:

This paper employs the vector autoregressive moving average method for validation. However, the experiment lacks comparative experiments, which fails to ensure its innovativeness. Additionally, there is a lack of ablation studies—for instance, what the results would be without the Johnson–Lindenstrauss (JL) projection method. Furthermore, the effectiveness of the reservoir size and spatial reuse is not sufficiently demonstrated.

Resubmission:

This paper addresses the quantum reservoir problem by proposing QuaRK, an end-to-end framework that couples a hardware-realistic quantum reservoir featurizer with a kernel-based readout scheme. The theoretical results are validated through experiments. However, more comprehensive experiments should be conducted to ensure the sufficiency of the experimental validation.

Questions And Suggestions For Rebuttal:

The overall quality of the paper is commendable; however, the experimental section lacks sufficiency. It is recommended to include comparative experiments and ablation studies.

Relevance: 2: Low - The connection to KDD is weak

Novelty: 3: Moderate - The paper introduces a new and interesting idea or approach that adds value to the field. The contribution is original and represents an advancement of existing knowledge, demonstrating solid innovation and creativity.

Technical Quality: 3: Moderate - The paper demonstrates solid technical quality with a sound methodology and thorough analysis. The results are reliable and well-supported. There may be minor issues, but they do not significantly undermine the overall quality. The work is competently executed and meets acceptable standards.

Presentation: 3: Moderate - The paper is organized and generally clear. The writing is mostly free of grammatical and typographical errors, making it easy to read. Figures and tables are effectively used to support the text. The presentation facilitates understanding and conveys the key points effectively.

Reproducibility: 3: Moderate - The paper provides a clear and detailed description of the methods, data, and procedures used. Supplementary materials, such as datasets and code, are available and sufficiently documented. Reproducing the results would be feasible with the provided information, though some effort may still be required.

Reviewer Confidence: 2: Low - The reviewer has some knowledge of the subject area and is somewhat familiar with the research methods and context of the paper. They understand the main points but may lack depth in certain areas. The reviewer is reasonably confident in their assessment but acknowledges some limitations in their expertise.

Ethics Review Flag: No

Ethics Review Description: no

Rebuttal by Authors

Rebuttal

by Authors (Abdallah Aaraba (/profile?id=~Abdallah_Aaraba1), Soumaya Cherkaoui (/profile?id=~Soumaya_Cherkaoui1), Ola Ahmad (/profile?id=~Ola_Ahmad1), Shengrui Wang (/profile?id=~Shengrui_Wang2))

📅 10 Apr 2026, 22:03

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Rebuttal:

We thank the reviewer for the constructive feedback. We agree that the original empirical section was too compact to make the validation roles sufficiently clear. We now separate them explicitly: the synthetic VARMA study is for QuaRK-only mechanism validation, while a separate real-world benchmark is for classical comparison under matched budget.

For the controlled VARMA split used in the paper ($w = 25$, $d = 3$, $N_{\text{tr}} = 5000$, $N_{\text{te}} = 1000$), we added ablations around the main design knobs. On the hardest Volterra task, the default exact setting gives test MSE 0.629388. Removing Johnson--Lindenstrauss (JL) gives 0.606072, which in this low-dimensional setting supports JL mainly as a scalability/compression mechanism rather than a universal accuracy booster. The architecture sweep also shows that the resource knobs matter: the best retuned qubit sweep reaches 0.478788 at $n = 6$; spatial reuse helps but not monotonically (exact $R = 8$ gives 0.5661); locality shows a trade-off, with $k = 1$ underfitting (0.7235) and $k = 3$ over-expanding the feature map (0.6593); and exact performance improves to 0.5941 at $\lambda_0 = 0.5$ before degrading again. These runs directly address the request for JL, reservoir size, and spatial reuse.

We also added a matched-budget temporal comparison against ESN (echo state network)+Matérn rather than mixing classical baselines into the synthetic ablation table. QuaRK is fixed to $n = 5$, $R = 3$, $k = 2$, $\lambda_0 = 0.5$, 3000 shots, with feature dimension $D = 315$; ESN+Matérn uses the same budget $D = 315$; both are evaluated over 5 seeds. Across 10 time-series extrinsic regression datasets spanning $w = 24 \dots 7500$ and $d = 1 \dots 728$, covering long univariate spectroscopy/gas-sensor trajectories, shorter multivariate tasks, and a hydraulic setting, QuaRK is better on 5/10. Wins include gas_sensor_array_ethanol (0.9579 ± 0.0263 vs 1.0446 ± 0.1064) and copper_concentration (1.0003 ± 0.0084 vs 1.0693 ± 0.1219), while live_fuel_moisture is a near-loss (0.9383 ± 0.0056 vs 0.9074 ± 0.0088).

So the intended claim remains restrained: not that QuaRK universally beats classical models, but that its design knobs have measurable effect in controlled ablations and that, under a matched temporal budget, it is competitive on several real datasets. For reproducibility, the added runs are tied to fixed splits, configs, seeds, and tabulated result artifacts prepared for publication upon acceptance.



➔ *Replying to Rebuttal by Authors*

Official Comment by Reviewer kc2J

Official Comment by Reviewer kc2J 📅 18 Apr 2026, 07:09

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

I am grateful to the authors for their detailed response explanations. As some concerns have been resolved, I will keep my current score unchanged.

Official Review of Submission3103 by Reviewer vKFL

Official Review by Reviewer vKFL 📅 23 Mar 2026, 12:30 (modified: 15 May 2026, 22:29)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer vKFL, Authors

📄 Revisions (/revisions?id=2DCacWIX9E)

Paper Summary:

This paper proposes and analyzes an end-to-end framework called QuaRK, which combines a hardware-realistic quantum reservoir feature extractor with a readout mechanism based on the Reproducing Kernel Hilbert Space (RKHS), and provides rigorous learning-theoretic guarantees of generalization for weakly correlated time series data.

Paper Strengths:

1. By innovatively introducing the Johnson-Lindenstrauss (JL) projection, this paper has effectively decoupled the quantum resource requirements from the dimension of the original input data, thereby enabling the processing of high-dimensional time series.
2. The quantum reservoir adopts a Contracted-Encoding Quantum Channel (CEQC) architecture, and mathematical proofs ensure that the model inherently possesses Echo State Property (ESP) and Fading Memory Property (FMP).
3. This paper provides detailed PAC learning theoretical bounds for β -mixed stationary processes, and establishes an explicit relationship between quantum resource allocation, measurement budget, and finite-sample performance.

Paper Weaknesses:

1. Empirical studies have been primarily limited to synthetic β -mixture VARMA tasks, lacking robust validation on complex real-world datasets with long-term memory or non-stationary characteristics.
2. Core performance analyses have been conducted mainly on idealized simulators, failing to fully quantify the impact of physical noise and finite-shot effects in real quantum hardware on actual generalization error.
3. Model performance is highly dependent on complex design choices such as the contraction factor, projection dimension, and kernel hyperparameters. No systematic approach for automatic parameter optimization has yet been provided.
4. For large-scale observation estimation, the optimal trade-off between measurement budget, observation locality k , and spatial multiplexing requires further experimental clarification.

Resubmission:

No.

Questions And Suggestions For Rebuttal:

1. Examine the generalization performance of the proposed framework beyond synthetic β -mixture processes, particularly its ability to handle real-world complex time series data characterized by non-stationarity or long-memory features.
2. Provide quantitative analysis of the specific impact of physical noise in real quantum hardware and finite-shot effects on theoretical generalization limits, and validate the model's performance in non-ideal environments.
3. Provide systematic selection guidelines or automated tuning schemes for key design variables, such as the contraction factor λ , the number of qubits n , and the Matérn kernel hyperparameters.
4. To clarify the performance boundaries under different resource configurations, please provide detailed experimental or theoretical explanations regarding the trade-offs between the measurement budget, the observation locality k , and the number of spatial multiplexing sub-reservoirs R .

Relevance: 3: Moderate - The work is somewhat relevant to the Research track of KDD and is of narrow interest to a sub-community

Novelty: 3: Moderate - The paper introduces a new and interesting idea or approach that adds value to the field. The contribution is original and represents an advancement of existing knowledge, demonstrating solid innovation and creativity.

Technical Quality: 2: Low - The paper has several technical weaknesses, such as minor methodological flaws, insufficient analysis, or unsupported conclusions. While the work shows some level of competence, it lacks thoroughness and precision. Improvements are necessary for it to be considered robust.

Presentation: 2: Low - The paper has noticeable issues with clarity and coherence. The writing may contain several grammatical and typographical errors. Figures and tables are present but may not be well-integrated or effectively used. The presentation allows for understanding but requires effort from the reader.

Reproducibility: 2: Low - The paper includes some information about the methods, data, and procedures, but key details are missing. There may be supplementary materials, but they are incomplete or unclear. Reproducing the results would require significant effort and additional information.

Reviewer Confidence: 1: Poor - The reviewer has limited knowledge of the subject area and is not very familiar with the specific research methods or context of the paper. The reviewer is unsure about their ability to accurately assess the paper and may have had difficulty understanding key aspects of the work. Their evaluation should be considered with caution.

Ethics Review Flag: No

Ethics Review Description: N/A



Rebuttal by Authors

Rebuttal

by Authors (Abdallah Aaraba (/profile?id=~Abdallah_Aaraba1), Soumaya Cherkaoui (/profile?id=~Soumaya_Cherkaoui1), Ola Ahmad (/profile?id=~Ola_Ahmad1), Shengrui Wang (/profile?id=~Shengrui_Wang2))

10 Apr 2026, 22:28

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Rebuttal:

We thank the reviewer for the detailed comments. We agree that the original version did not quantify practical effects enough. The main contribution remains the QuaRK methodology and dependent-data theory; the rebuttal additions are meant to calibrate resource knobs empirically.

For non-ideal measurement effects, we added a finite-shot sweep on the paper VARMA Volterra split ($w = 25, d = 3, N_{\text{tr}} = 5000, N_{\text{te}} = 1000$). Test MSE degrades from the exact baseline 0.629388 to 1.047676 at 100 shots, then improves to 0.834551 at 1000 and 0.701464 at 10000. This does not replace a full hardware-noise study, but it directly quantifies the finite-shot component of non-ideal behavior. In the pilot, 3000 shots was selected since its mean deviation from exact is 0.0122 and its max deviation is 0.0644.

For tuning guidance, we extended the QuaRK-only VARMA architecture sweep. On the hardest Volterra task, the best retuned qubit-count setting reaches 0.478788 at $n = 6$. The same sweep clarifies other trade-offs: exact $R = 8$ gives 0.5661; changing locality away from the default $k = 2$ did not help on this split ($k = 1$: 0.7235, $k = 3$: 0.6593); and exact performance improves from 0.629388 to 0.5941 at $\lambda_0 = 0.5$. We present these as operating-point guidance rather than an automatic tuning scheme.

Beyond synthetic β -mixing VARMA, we also added a matched-budget benchmark against ESN (echo state network)+Matérn readout, our fairest matched control, isolating the reservoir while keeping the readout matched. QuaRK is fixed to $n = 5, R = 3, k = 2, \lambda_0 = 0.5$, 3000 shots, with feature dimension $D = 315$; ESN+Matérn uses the same $D = 315$; both are averaged over 5 seeds. Across 10 time series extrinsic regression (TSER) datasets spanning $w = 24 \dots 7500$ and $d = 1 \dots 728$, QuaRK is better on 5/10; for example, gas_sensor_array_ethanol gives 0.9579 ± 0.0263 vs 1.0446 ± 0.1064 , while live_fuel_moisture is a near-loss at 0.9383 ± 0.0056 vs 0.9074 ± 0.0088 .

We do not claim that this fully resolves hardware-noise validation or establishes universal superiority over classical baselines. Rather, the new evidence quantifies finite-shot effects, makes the roles of n, R, k , and λ more explicit, and shows that QuaRK is competitive on several real datasets under a matched temporal budget. For reproducibility, all added results are tied to fixed configs, seeds, splits, and tabulated artifacts.



Replying to Rebuttal by Authors

Official Comment by Reviewer vKFL

Official Comment by Reviewer vKFL 18 Apr 2026, 04:08

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

I thank the authors for the detailed rebuttal and the additional clarifications. The new empirical results are helpful in addressing some of my concerns, particularly regarding non-ideal settings. However, I still find that the validation on real-world datasets remains somewhat limited and it does not sufficiently resolve my main concerns. Therefore, I will maintain my current score.

Official Review of Submission3103 by Reviewer MSDS

Official Review by Reviewer MSDS 19 Mar 2026, 09:43 (modified: 15 May 2026, 22:29)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer MSDS, Authors

Paper Summary:

QuaRK combines a contractive quantum reservoir with a Matérn kernel ridge regression readout for time-series learning. A JL projection decouples qubit count from input dimension, classical shadows provide efficient feature extraction via k -local observables, and spatial multiplexing across sub-reservoirs increases expressivity. Theoretically, the paper establishes an interpolation guarantee under logarithmic qubit scaling (Theorem 1) and a generalization bound for β -mixing data with three interpretable terms: Rademacher complexity, mixing penalty, and exponentially decaying fading-memory residual (Theorem 2). Synthetic VARMA experiments confirm both predictions qualitatively.

Paper Strengths:

- The paper addresses an interesting and relatively underexplored intersection of quantum reservoir computing, kernel methods, and temporal learning.
- The overall pipeline is coherent and well-motivated: JL projection, contractive reservoir dynamics, classical-shadow measurements, and kernel ridge regression fit together cleanly.
- The learning-theoretic analysis is a meaningful contribution. In particular, the generalization result for dependent β -mixing temporal data.
- The presentation is generally clear. The method section is organized, and Figures 1–2 help explain the architecture.
- Appendix B provides useful detail on the synthetic data generation process and target functionals, which improves transparency.

Paper Weaknesses:

All experiments are on synthetic VARMA-generated data in a single low-dimensional setting ($d=3$, $n=5$ per sub-reservoir), across three synthetic target functionals. No real-world time series and no classical baselines (echo state networks, linear ridge regression, random features) are included. Without these, it is unclear whether the quantum reservoir adds value over the kernel readout alone.

The experiments also mainly validate qualitative trends predicted by the theory, but they do not demonstrate that the proposed method is competitive or that the quantum component provides measurable benefit. A central practical claim is scalability to high-dimensional data through JL projection, yet the reported experiments use only $d=3$ and $n=5$. This size, however, is too small to substantiate the scalability claim empirically.

Beyond sweeps over regularization and training-set size, the paper lacks ablations and robustness checks. There is no sensitivity analysis over hyperparameters like the number of sub-reservoirs R , qubit count n , observable family/locality, or contraction values. No error bars or confidence intervals are reported.

Finally, although the paper emphasizes hardware realism, the validation is entirely simulation-based. The analysis also does not fully incorporate realistic hardware noise or finite-shot estimation error, which is important because the method relies on classical-shadow measurements and a nontrivial reset/ancilla mechanism.

Resubmission:

N/A

Questions And Suggestions For Rebuttal:

1. Could you provide results on real-world time-series benchmarks and compare against classical baselines such as an echo state network, linear ridge regression on raw windowed features, and a classical random-feature kernel with the same output dimension as the quantum featurizer?
2. Theorem 1 essentially relies on injective features plus a positive definite kernel on the training set. Could you provide an ablation that isolates the quantum reservoir's contribution, for instance by replacing it with a classical random nonlinear map of the same feature dimension and comparing interpolation and generalization behavior?
3. How sensitive are the reported results to reservoir initialization, JL projection randomness, and data-generation randomness? Please report averages and standard deviations (or confidence intervals) over multiple runs.
4. How sensitive is performance to qubit count n , the contraction factors λ_r , the shot budget, and the input dimension d ? A systematic ablation over these parameters would also clarify how the experimental choice $n=5$ relates to the theoretical JL-style scaling regime.

Relevance: 3: Moderate - The work is somewhat relevant to the Research track of KDD and is of narrow interest to a sub-community

Novelty: 3: Moderate - The paper introduces a new and interesting idea or approach that adds value to the field. The contribution is original and represents an advancement of existing knowledge, demonstrating solid innovation and creativity.

Technical Quality: 3: Moderate - The paper demonstrates solid technical quality with a sound methodology and thorough analysis. The results are reliable and well-supported. There may be minor issues, but they do not significantly undermine the overall quality. The work is competently executed and meets acceptable standards.

Presentation: 3: Moderate - The paper is organized and generally clear. The writing is mostly free of grammatical and typographical errors, making it easy to read. Figures and tables are effectively used to support the text. The presentation facilitates understanding and conveys the key points effectively.

Reproducibility: 2: Low - The paper includes some information about the methods, data, and procedures, but key details are missing. There may be supplementary materials, but they are incomplete or unclear. Reproducing the results would require significant effort and additional information.

Reviewer Confidence: 3: Moderate - The reviewer has a good understanding of the subject area and is familiar with the research methods and context of the paper. They feel confident in their ability to accurately assess the quality and significance of the work. Their evaluation is based on a solid grasp of the content and context.

Ethics Review Flag: No

Ethics Review Description: /



Rebuttal by Authors

Rebuttal

by Authors ( Abdallah Aaraba (/profile?id=~Abdallah_Aaraba1), Soumaya Cherkaoui (/profile?id=~Soumaya_Cherkaoui1), Ola Ahmad (/profile?id=~Ola_Ahmad1), Shengrui Wang (/profile?id=~Shengrui_Wang2))

 10 Apr 2026, 11:52 (modified: 10 Apr 2026, 21:51)

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

 Revisions (/revisions?id=kOrdib9EZ2)

Rebuttal:

Thank you for the careful, technically precise review. We agree that the original empirical section was too limited for practical calibration. The primary contribution is the QuaRK methodology and dependent-data theory; the original VARMA study was designed to validate predicted interpolation/generalization behavior and resource knobs, not blanket superiority over classical temporal models.

To address this, we added two targeted rebuttal packages. First, on the same fixed VARMA split as the paper ($w = 25$, $d = 3$, $N_{tr} = 5000$, $N_{te} = 1000$), we ran controlled QuaRK-only mechanism ablations. On the hardest Volterra task, the default exact setting gives test MSE 0.629388. The resource knobs materially affect performance: the best retuned qubit sweep reaches 0.478788 at $n = 6$; spatial reuse helps but not monotonically (exact $R = 8$ gives 0.5661); and locality shows a trade-off, with $k = 1$ underfitting (0.7235) and $k = 3$ over-expanding the feature map (0.6593). The contraction factor changes the effective memory scale, with exact performance improving to 0.5941 at $\lambda_0 = 0.5$ before degrading again at larger values. We also quantified finite-shot degradation on the same task: $100 \rightarrow 1.047676$, $1000 \rightarrow 0.834551$, and $10000 \rightarrow 0.701464$.

Second, for external calibration, we added a matched-budget real-world temporal benchmark against ESN (echo state network)+Matérn, our fairest temporal control because the readout is matched. QuaRK is fixed to $n = 5$, $R = 3$, $k = 2$, $D = 315$, $\lambda_0 = 0.5$, and 3000 shots; ESN+Matérn uses the same budget $D = 315$; both are averaged over 5 seeds. Across 10 time-series extrinsic regression (TSER) datasets spanning $w = 24 \dots 7500$ and $d = 1 \dots 728$, QuaRK is better on 5/10. Representative wins are gas_sensor_array_ethanol dataset (0.9579 ± 0.0263 vs 1.0446 ± 0.1064) and copper_concentration dataset (1.0003 ± 0.0084 vs 1.0693 ± 0.1219), while live_fuel_moisture dataset is a near-loss (0.9383 ± 0.0056 vs 0.9074 ± 0.0088). We do not present this as universal superiority; rather, it shows QuaRK can be competitive under a matched temporal budget while preserving the theory-first scope.

Regarding Theorem~1, we agree its role is interpolation feasibility under injective features plus a positive definite kernel, not a quantum-advantage claim. For reproducibility, all added results are tied to fixed configs, seeds, and dataset manifests prepared for release.



➔ *Replying to Rebuttal by Authors*

Official Comment by Reviewer MSDS

Official Comment by Reviewer MSDS 📅 16 Apr 2026, 07:29

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

Thank you for the rebuttal and taking the time to provide all these additional experiment results. I think the
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